

Asher Siddiqui

Karachi, Pakistan | +92-3302463301 | ashersiddique127@gmail.com | [LinkedIn](#) | [Github](#)

EDUCATION

FAST - NUCES

Bachelor of Science in Software Engineering

Karachi, Pakistan

Expected Graduation: June 2025

Relevant Courses: Applied DevOps, Cloud Security, Agile Software Project Management

Awards & Honors: Dean's List of Honors (3.5+ SGPA) in Fall 2023, Spring 2024, Fall 2024

TECHNICAL SKILLS

Languages: Go, Python, Java, C

Backend & Frameworks: Echo, Spring Boot, Postman

Databases: MySQL, Oracle RDBMS, MongoDB, Redis

DevOps & Cloud: Docker, Docker Compose, Kubernetes, AWS, Linux, Bash, Ansible, Prometheus, Node Exporter, Falco

EXPERIENCE

TAFSOL Technologies

Mar 2025 – Present

Software Engineering Intern

Karachi, Pakistan (Hybrid)

- Enabled security auditing and compliance by implementing a logging system with size based rotation for security and error logs, improving visibility and optimizing storage consumption.
- Secured user's personal and financial data by integrating AES-CBC with HMAC for persistent storage, ensuring confidentiality and integrity of sensitive information.
- Streamlined testing workflows by deploying Docker Compose builds for test environments, reducing setup time and ensuring consistent testing conditions across development teams.
- Reduced manual steps for funds transfer by developing a Quick Pay feature with QR code generation, cutting user transaction time by 60%.
- Reduced on premise storage and hardware costs by integrating AWS S3 using AWS SDK, enabling encrypted image storage and controlled access.
- Improved transparency by developing a transaction history API that provided users with access to their financial activities, enhancing user trust.
- Improved API reliability by conducting functional testing with Postman, validating responses and identifying issues early.

Grayhat Private Limited

July 2024 – Sept 2024

Software Engineering Intern

Islamabad, Pakistan (Remote)

- Developed and managed the MVP for v0.dev's VS Code extension, coordinating with one other developer to accelerate delivery and increase user satisfaction.
- Built an automated developer productivity integration system, synchronizing events across GitHub, Kimai, Activity Watch, and Plane, enabling streamlined workflow tracking and increased developer efficiency.

PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | *Kubernetes, Ansible, Prometheus, Falco, Go, Python* | [Github](#)

- Deployed a 3-node botnet and a 4-node bare-metal Kubernetes cluster using Ansible to simulate real DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline to log physical resource-related system calls and usage metrics, integrating Falco and Node Exporter, enabling seamless training data generation.
- Developed and deployed attack simulation scripts using Python on the C&C server and the botnet infrastructure, enabling orchestration of controlled DDoS attacks.

Rate Limiter | *Go, Docker, Redis* | [Github](#)

- Engineered a Layer 4 rate limiter supporting Token Bucket and Sliding Window algorithms using Go, preventing API misuse through uncontrolled requests.
- Optimized HTTP responses by adding custom rate-limiting headers, improving client-side request handling and ensuring request quota transparency.

- Improved system scalability and response by leveraging Redis for state management, ensuring low-latency request processing under high traffic.

Load Balancer | *Go, Express.js, Docker, Git, HTML* | [Github](#)

- Designed and implemented a high-throughput HTTP reverse proxy implementing Round Robin and Least Connection algorithms using Go, achieving 2000 requests per second with minimal latency.
- Increased reliability by integrating health checks, ensuring automatic rerouting of traffic to available servers.
- Containerized the load balancer using Docker, enabling seamless deployment and scalability, and consistent performance across environments.

CUDA-based Mandelbrot Set Renderer | *CUDA, C, Python* | [Github](#)

- Developed a GPU accelerated Mandelbrot Set renderer using CUDA C, enabling highly parallel computation for fractal generation and achieving an 82% speed improvement over CPU based rendering.
- Implemented color mapping for detailed visualization of the set's complex structure.